In this journal entry, I will share my thoughts about module 5 and fixing datasets for machine learning. After learning how to do the preprocessing of the machine learning dataset, my main takeaway is how to deal with different situations that may occur when it comes to the preprocessing of machine learning datasets. Using the same Titanic dataset from module 4, it helped me see more problems with raw data and how some of these may happen in real time.

Before the lab started, I thought that data preparation was just about removing anything incomplete. What I didn't expect was the amount of estimation and educated guessing while doing the preparation. When I was working with the age and Cabin columns, it showed that more than half of them were empty. This made me think that the best solution to fix the problem is just to delete them. But the lab addressed this as not the best solution and rather wanted me to impute the data with values that were guessed. At first, this came as a surprise to me, as to me, this could throw off the AI because if the information is guessed, then it may lead to bad AI. However, when I worked through task 1, I realized that assuming is better than nothing. As if you want to delete that row and other information that may be useful for later, like the fare, is gone as well. I also realized that the guesses were based on the dataset instead of outside information. As the median may not be exactly the age of the passenger, it will be better than losing the other parts of the data.

This realization came during task one, when I was asked to fill in the Age column and was told to use the median instead of the mean. At first, I questioned the decision, as the mean is used for most averages of values in a list and is quite accurate. However, as I wrote code for both the median and the mean, I realized that the median is much better at making sure outliers don't affect the data. Elderly passengers can pull the mean up, while the median is the true middle. In this dataset, the difference was only 1.7, which shows that even though it may not make a huge difference, there still is one.

After working with missing values, the next problem I faced was that instead of values not being there, they were in the wrong format. In the sex column. If we gave this information to an AI, it could mislead it in the wrong direction. The first thought in mind was to just give each category a number. However, the lab showed the one-hot encoding method. This method, I noticed, was like a binary system to display the categories, where 1 means that it's true and 0 means that it's not true. When looking over the code, I noticed that we were deleting a column, but after thinking about it makes logical sense as if Sex_Male is false, then we know they are a female. However, I still wanted to know why just assigning a number to each category wasn't a good solution, as this method is called Ordinal Encoding, where each category gets an ascending order number. However, after I investigated it more, I realized that this almost always throws the AI off, as the AI will take the higher number as being better than the other, but it could just be different colors, and one is not better than the other.

Feature scaling is something I have never heard before, but I now realize that it's great for making sure the AI isn't dominated by one feature and making sure everything is equal. I saw the age and fare go from 22 and 71 down to -0.56 and 0.78. I thought the data was being broken down and was less important, but I now understand that the standard scaler was taking the age, subtracting it from the mean, and then dividing by the standard deviation, so I now know that this is not a mistake and the passenger was just below average. The reason for these matters as fare can go up 512 while the age only goes up to 80. If you keep them like they are, the AI will pay more attention to the much higher number.

Module 5 showed me that even though making the model may be hard, the prep is much more time consuming than any other. Every step you take to make the data for the machine is a tedious process that requires careful consideration, as this part can affect the rest of your time building the model, and the better your data preparation, the better your machine can become.